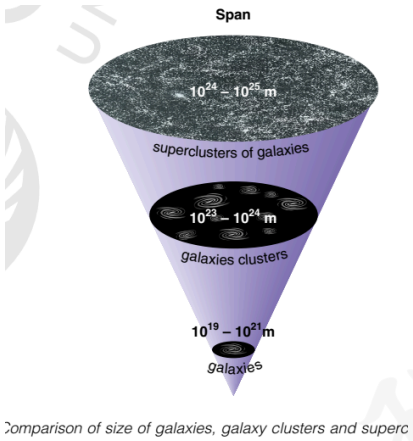
E1.1.1 Hierarchy and scales of celestial systems

Stars and planetary systems lie within galaxies. Galaxies form groups, clusters and superclusters. Star clusters are groups of stars. Nebulae consist mainly of gas and dust. Both are usually found within galaxies. Compare sizes using the same units. Size ranges of different objects may overlap.



Comparison of size of galaxies, galaxy clusters and superc

Compare the size hierarchy of galaxies, galaxy clusters and superclusters.

E1.1.2 The astronomical unit

The astronomical unit, AU, is a length unit suited to Solar System distances, approximately the mean Earth-Sun distance or 1.50×10^{11} m. Convert AU, metres and other units consistently before comparing distances.

$$1\ \text{AU} \approx 1.50 \times 10^{11}\ \text{m}$$

One astronomical unit is approximately the mean Earth–Sun distance. Textbook rounded conversion for Solar System distances.

E1.1.3 The light-year

A light-year is the distance light travels in vacuum in one year, approximately 9.46×10^{15} m. It is a length, not a time. Calculate it from light speed times the number of seconds in a year. A star's distance in light-years indicates approximate light-travel time in years.

$$1\ \text{ly} \approx 9.46 \times 10^{15}\ \text{m}$$

A light-year is the distance travelled by light in vacuum in one year; a unit of distance, not time.

E1.1.4 The geometrical meaning of a parsec

A parsec is the distance corresponding to an annual parallax of one arcsecond. The geometrical baseline is 1 AU, giving approximately 206265 AU or 3.26 light-years by the small-angle relation. An arcsecond measures angle, not time. A parsec measures distance despite the historical reference to seconds in its name.

$$1\ \text{pc} \approx 3.09 \times 10^{16}\ \text{m} \approx 3.26\ \text{ly}$$
$$1\ \text{pc} \approx 206\,265\ \text{AU}$$

Parsec: distance corresponding to annual parallax of one arcsecond. Rounded conversions.

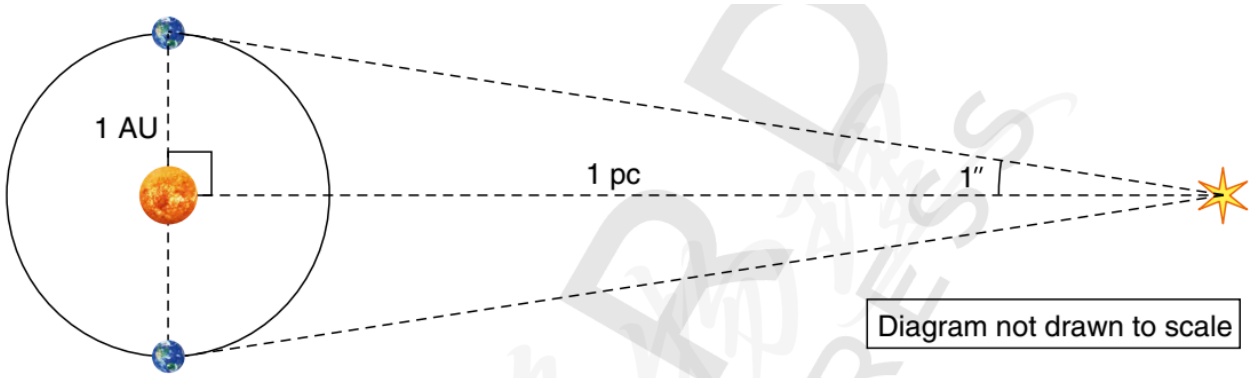


Fig 3.1i *Definition of one parsec.*

The one-AU baseline subtends the annual parallax angle. The full apparent shift over six months is twice that angle.

E1.1.5 Retrograde planetary motion

Retrograde motion is a planet's temporary apparent westward motion relative to background stars. In the heliocentric model, different orbital speeds change viewing geometry. For example, an outer planet can appear retrograde as Earth overtakes it, while the planet's actual orbital direction stays unchanged.

Ptolemy's geocentric model uses an epicycle and deferent. The planet travels around a small circle, the epicycle, whose centre travels around a larger circle about Earth, the deferent. Combining these motions can also produce an apparent retrograde path.

E1.1.6 Earth rotation, the celestial sphere and latitude

Earth rotates from west to east, making celestial objects appear to move east to west each day. Imagine distant objects on a celestial sphere centred on the observer. Earth's extended axis points towards the celestial poles, about which daily apparent motion occurs.

The angle between the local horizon and Earth's axis changes with latitude, so the same object's rising and setting path depends on the observer's location. In the northern hemisphere, the north celestial pole's altitude equals the observer's latitude. Near the North Pole, paths run nearly parallel to the horizon; near the equator, rising and setting paths are closer to perpendicular.

Past-paper application: at the Sun's upper meridian passage, its altitude is $h = 90^\circ - |\varphi - \delta|$. Here φ is latitude and δ is solar declination: approximately 0° at the equinoxes, +23.5° at the June solstice and −23.5° at the December solstice.

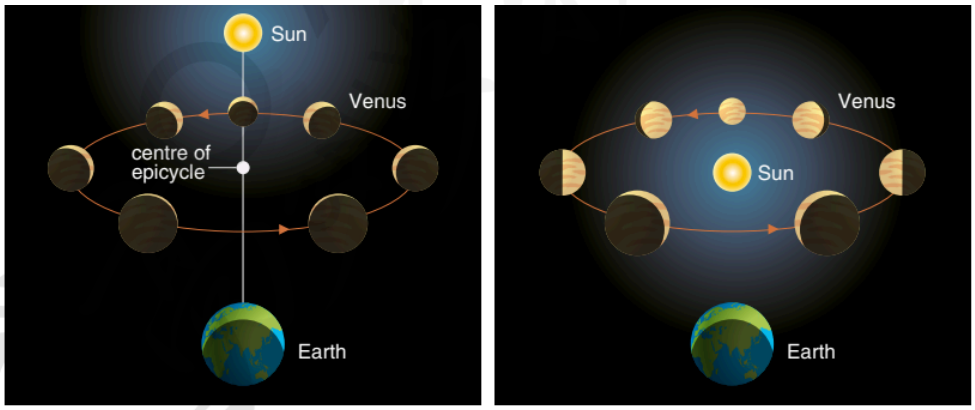
E1.1.7 Phases of Venus and Jupiter's moons

Venus shows phases from crescent to nearly full. It is nearer and has a larger apparent diameter when crescent-shaped, and farther with a smaller diameter when nearly full. This fits an orbit around the Sun, but not the Ptolemaic arrangement placing Venus always between Earth and the Sun. Jupiter's moons show that some celestial bodies orbit objects other than Earth.



Fig 1.4h *Different phases of Venus over a period of time. The planet looks bigger when it*

Venus changes apparent diameter as its phase changes.



(i) In the Ptolemaic model, Venue does not go *(ii) In the heliocentric model, Venus shows*

Compare the phases of Venus predicted by the Ptolemaic and heliocentric models.